



How Does the Food Industry Set Prices?

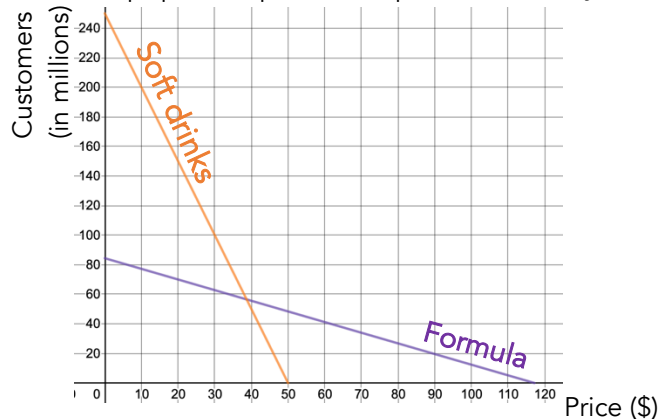
The food industry is made up of a collective of diverse businesses that together make the majority of the food consumed by the world's population. How do these businesses decide which items to advertise and how much to charge for each product?

1. What types of food or drink do you see advertised often? For what types do you see very little advertising? Why do you think this is?

Often: Coca-cola, chips, smoothies
 Less often: fruit, bread, eggs, milk

They don't need to advertise things people naturally buy + need.

2. The lines below model the possible relationship between the price and number of customers for two popular supermarket purchases: baby formula and soft drinks.



- a) What do you notice about the two graphs? What do you wonder?

The market for soft drinks is much bigger but people are willing to pay more for formula. I wonder how companies figure out this kind of data.

- b) Which customer base, formula or soft drinks, is more swayed by an increase in price? How do you know?

Soft drink consumers are more likely to decrease consumption/stop buying pop when price increases. The slope is steeper so the change in customers per price increase is higher.

3. Look at the data in the table below.

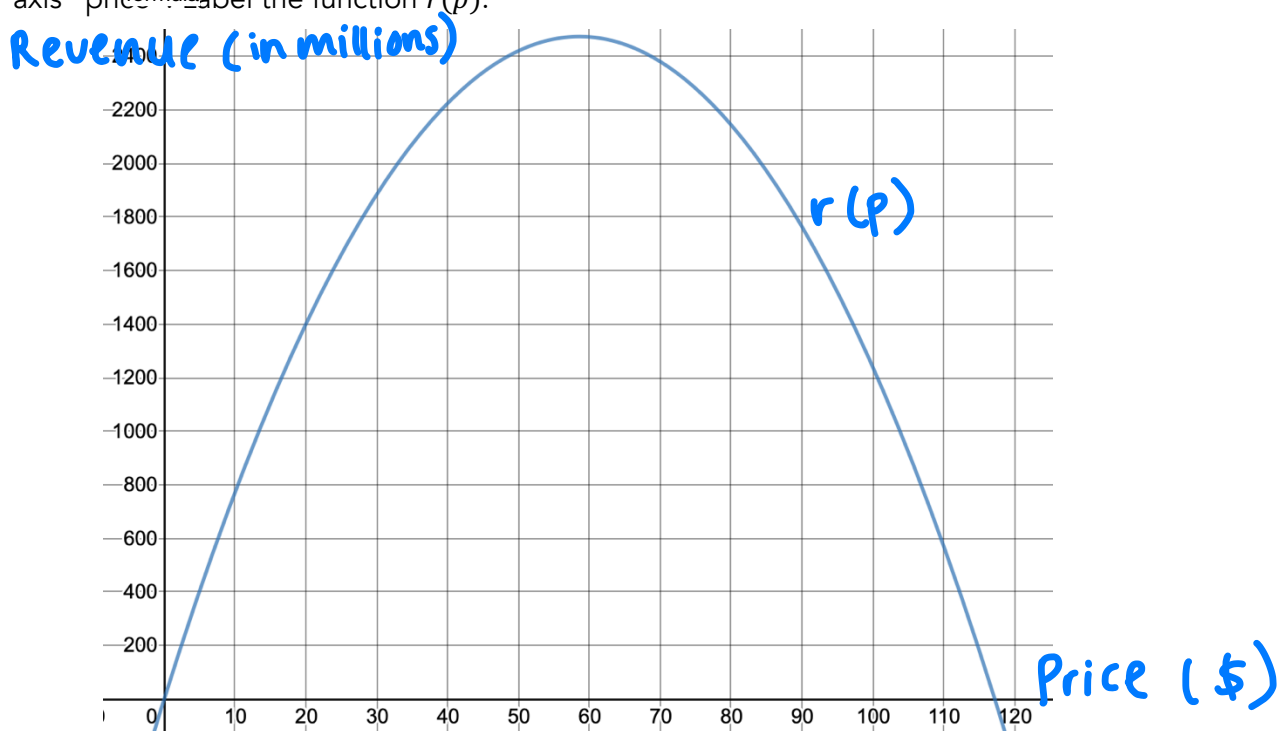
- a. Calculate the company's **revenue** for each item when the price is \$10 and when the price is \$45.

	Price=\$10		Price=\$45	
	Customers	Revenue	Customers	Revenue
Soft Drinks	200 million	2,000 million	30 million	1,350 million
Formula	78 million	780 million	52 million	2,340 million

- b. Do you think it's a good idea for food companies to always charge more? Explain.

No. If you overcharge you will lose so many customers that you end up making less money.

4. The graph below gives the food company's revenue for baby formula. The x-axis represents the price of the formula. Label the y-axis "revenue (in millions)" and the x-axis "price (in dollars)". Label the function $r(p)$.



5. a. The new intern at the company suggests selling the formula for \$20. Is this the best price? How do you know?

$r(p)$ is increasing on $(0, 60)$

No, because you could charge more + make more.

- b. The second intern suggests selling the formula for \$80. Is this the best price? How do you know?

$r(p)$ is decreasing on $(60, 117)$

No, this is too expensive. By charging less you can actually make more money.

- c. If the goal is to make the most money, what is the optimal price for the formula? How do you know?

Roughly \$60, because revenue is at its peak. Relative maximum is 2500 (million) \$ in revenue.

6. At what price(s) will the company make no money on formula? Why?

set $r(p) = 0$ to find x-intercepts.

\$0 and roughly \$117. If you offer it for free you don't make money but at \$117, nobody buys it.

7. Between $p = 0$ and $p = 60$, how much does revenue change, on average, when the price of formula increases by \$1? Explain.

Average rate of change: $\frac{r(60) - r(0)}{60 - 0}$

$$\frac{2500}{60} = 41.67 \text{ (million)}$$

For every \$1 increase in price, the revenue increases by 41.67 million

8. When or why might companies use factors besides revenue to determine their prices?

For items like baby formula, maximizing revenue might not be the best because if families can't afford it there are serious consequences.

dollars, on average

Lesson 1.3 – Rates of Change and Graph Behavior

QuickNotes

A function is increasing on (a,b) if y-values are getting bigger whenever $a < x < b$

A function is decreasing on (a,b) if y-values are getting smaller whenever $a < x < b$.

Relative maxima/minima are the y-values at which the function has high points + low points.

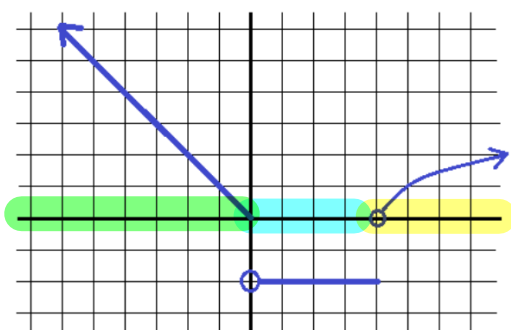
To find zeros/x-intercepts of a function, set $f(x) = 0$ and solve.

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

"On average, the y-variable increases/decreases by slope per unit of x-variable"

Check Your Understanding

1. Give the intervals on which $f(x)$ is increasing, decreasing, and constant.



Increasing on $(4, \infty)$
Decreasing on $(-\infty, 0)$
Constant on $[0, 4]$

2. The function $g(x)$ is shown below.
 - a. Find all x-values where $g(x) = 0$.

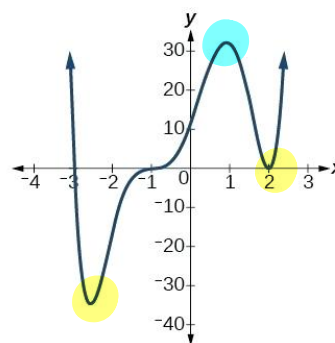
$$x = -3, -1, 2$$

- b. Estimate the relative minimums of $g(x)$.

$$-35, 0$$

- c. Estimate the relative maximums of $g(x)$.

$$32$$



3. For $g(x) = x^2 - 5x + 8$, find the average rate of change on the interval $[1, 6]$.

$$\begin{aligned} \frac{g(6) - g(1)}{6 - 1} &= \frac{(6^2 - 5(6) + 8) - (1^2 - 5(1) + 8)}{6 - 1} \\ &= \frac{14 - 4}{5} = \frac{10}{5} = 2 \end{aligned}$$